

Li Mingquan



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Date of birth: 2004.10.12 Hometown: Wuhan,Hubei

Education

2023-present Lanzhou University (985 / Top China University)
School of Physics Microelectronics Science and Engineering (Top 30%) Bachelor

Awards

National College Student Computer System Capability Competition – CPU Design Competition			
	Excellence Award	National Level	2026.8
Fuwei Cup National College Student Electronic Design Competition			
	National Third Prize	National Level	2026.8

Project Experience

1. 32-bit Five-Stage Pipelined MIPS Processor Sole Developer Mar.–Aug. 2026

- Tech Stack:** Verilog / Vivado / MIPS
- **Overview:** Independently designed a **32-bit five-stage pipelined MIPS** processor supporting precise exception and interrupt handling, implementing **47 MIPS instructions** and building a minimal SoC system with **hazard handling, a CP0 exception handling module, I-Cache, and UART peripherals.**
- Key Challenges:**
- Optimized the multiplier/divider and Load-Store forwarding on critical pipeline paths, and introduced mechanisms including I-Cache and non-blocking write buffers, increasing the maximum clock frequency from **75 MHz to 117 MHz.**
 - Optimized critical paths in I-Cache, pipeline control, and comparison logic through **Tag/Valid array optimization, parallel comparators, expanded Stall generation, EqualD comparator sharing, two-stage PC enable decomposition, and fanout replication.**
- Performance:** IPC 0.67; maximum synthesized frequency: 117 MHz

2. Intelligent Digital Security Sensor Core Developer Jun.–Aug. 2026

- Tech Stack:** FPGA/Verilog / Vivado / APB Bus / CDC Clock Domain Crossing Design
- **Overview:** Independently designed a fully digital autonomous hardware defense subsystem against fault injection attacks. The architecture consists of a **sensing layer** (temperature/voltage and clock-glitch sensors), a **decision layer** (including a watchdog and central decision system), and a **configuration layer** (APB register interface). The system adopts a four-clock-domain architecture and a fully digital synthesizable design, enabling hierarchical detection, decision-making, and autonomous response to threats ranging from **nanosecond transient glitches to slow attacks.**
- Key Challenges:**
- Introduced virtual-clock strong driving, a 16-level LUT delay-line (TDL) glitch sensor, automatic calibration, and a saturation-counter debounce arbitration mechanism to improve glitch sensing accuracy and threat-level classification.
 - Designed glitch-injection path pass-through, cross-domain buffer stages, and startup-edge filtering mechanisms to address multi-clock-domain timing issues, optimize CDC timing, and eliminate setup-time violations.
- Performance:** 1382 LUT / 2006 FF / Power: 0.223 W / Maximum clock frequency: 200 MHz

Professional Skills

- Proficient in C, Python, and Verilog.
- Proficient in FPGA development and EDA tools.